F78AA1 – Actuarial & Financial Mathematics

Feedback on Dec 2010 Exam

The following comments cover most of the features that emerged consistently in the 2010 exam. The examination paper and solution are available on Vision.

1. Parts (a) and (b) required the straightforward calculation of the equivalent effective interest rate over a different time period and was generally well attempted.

   Part (c) required two specific cash flows to be accumulated to both the end of the year and discounted to the start of the year and was also generally well attempted. Where marks were lost, it was due to either the careless calculation of the number of months or the effective accumulation / discount factor for the relevant period.

2. This question was a straightforward application of changing the terms of a mortgage albeit framed in a specific scenario.

   Part (a) was universally performed correctly.

   Part (b) required the derivation of three formulae relevant to the loan schedule and was less well attempted with very few correct derivations. The most common approach involved merely stating the actual formulae. Half marks were generously awarded if the formulae were correct with no marks being awarded for incorrect formulae.

   Part(c) was generally well attempted. Where marks were lost, it was mainly due to an error in the calculation of the capital outstanding either by not allowing for the four years during which no interest payments were made or by mis-calculating the outstanding term of the new loan.

   Part (d) was also well attempted except in the calculation of the outstanding term of the new loan as above.

3. Part (a) and (b) involved straightforward calculations involving TWRR & MWRR. In part (b), many students unnecessarily wasted time using linear interpolation to demonstrate that the return was 1.77% instead of just evaluating the expression at 1.77%. Part (c) was generally well attempted even if the answers were not well expressed.

4. In part (a), it was only required to state that **both** the interest and capital repayments were linked to the RPI index and that they were attractive to investors who sought a real rate of return. Many students wrote at length about the economy, or made generalisations that RPI gilts provided a better return than conventional gilts i.e. that there was a risk free arbitrage!

Part (b) was a very straightforward question, requiring the calculation of two payments due under the indexed linked gilt, and was generally well attempted. Some students failed to understand what was being asked and wrote at length the equation of value for the index-linked gilt.

Part(c) seemed to cause some unnecessary confusion and it appeared that some students were just guessing. In short, if an investor is interested in a certain monetary yield he will choose the conventional gilt as the returns under an index-linked gilt are unknown and dependant on future inflation.

5. This was a relatively straightforward question on bonds. However, for some reason about 20% of students carelessly assumed incorrectly that the term of the bond was actually 25 years!

   Part (a) was well attempted. However, marks were lost if the answer did not include some calculation indicating that CGT was actually payable.

   Part (b) was also well attempted with most students being able to re-arrange the equation correctly. Those students who had calculated the wrong price in part (a) were not further penalised.

6. This question combined straightforward bookwork together with the latter part requiring an understanding of the (geometric series) procedures in deriving annuity formulae.

   Part (a) was very straightforward and was well attempted in general. However, it was disappointing that some students showed so little understanding of the basics of the course.

   Part (b) was either performed extremely well or extremely poorly. Those students who had a good understanding of the procedure involving geometric series managed to work through the algebra and derive the equations. There were several cases of some very clear thinking in deriving the formulae.

   Part(c) should have been relatively straightforward utilising the results in part (a) and part (b). However, most students who managed to answer the question started from first principles again as opposed to using the results already derived.

7. This question, which was dealt with explicitly in lectures and in a spreadsheet uploaded to Vision, was very poorly attempted even though it dealt only with the fundamentals of the course. Marks were lost in not being able to interpret correctly the cash flow information for each of the lenders. Part (b) was very poorly performed with very few students being able to calculate the effective cost of borrowing - which included the fixed expenses - **to the investor**.